

D421

Discrete Math: Functions and Relations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A campus IT team surveyed 200 laptops: 120 run antivirus A, 90 have disk encryption, and 45 have both. The survey was run over two weeks by student workers. How many laptops have at least one of the two protections?

- A. 35
- B. 200
- C. 210
- D. 165

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Sets A and B have $|A| = 40$, $|B| = 35$, and $|A \cup B| = 60$. What is $|A \cap B|$?

- A. 75
- B. 135
- C. 25
- D. 15

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A configuration tool offers 6 optional features that can be turned on in any mix, and the vendor charges per seat. How many different feature sets are possible, counting none and all?

- A. 36
- B. 64
- C. 720
- D. 63

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A university gives each of its 30,000 students a unique ID drawn from 1,000,000 possible IDs. The ID system was built in 2012. How is the function f : Students $\rightarrow$ IDs best described?

- A. Injective but not surjective.
- B. Not a function, because some IDs are unused.
- C. Surjective but not injective.
- D. Bijective.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On the set of devices in a network inventory, define R by “ x is on the same VLAN as y .” The inventory lists 812 devices from four vendors. What kind of relation is R ?

- A. An equivalence relation.
- B. Symmetric but not transitive.
- C. A partial order.
- D. Not a relation, since the vendors differ.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

The relation $\leq$ on the integers is reflexive, antisymmetric, and transitive. What kind of relation is it?

- A. An equivalence relation.
- B. A function.
- C. A partial order (in fact a total order).
- D. Not a valid relation, since every relation must be symmetric.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A help desk closes 40 tickets in week 1 and closes 6 more each week than the week before as new staff ramp up. The manager also tracks satisfaction scores averaging 4.3. How many tickets does it close in week 10?

- A. 670
- B. 100
- C. 94
- D. 43

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

An arithmetic sequence has first term 7 and common difference 5. Which term number equals 102?

- A. 102 is not a term of the sequence.
- B. 20
- C. 95
- D. 19

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Backup storage use starts at 2 TB in year 1 and doubles each year; every year's backups are kept. The array uses RAID 6 across 12 disks. How much storage is used in total across years 1 through 5?

- A. 64 TB
- B. 10 TB
- C. 32 TB
- D. 62 TB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A geometric sequence has $a_1 = 3$ and $a_4 = 81$. What is the common ratio?

- A. 3
- B. 27
- C. 26
- D. 9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A script loops i from 1 to 100 and adds i to a counter on each pass; it runs on a server with 16 cores. What is the final counter value?

- A. 5,000
- B. 5,050
- C. 10,000
- D. 1,600

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a data-transform pipeline, $f(x) = 2x + 3$ and $g(x) = x^2$; the pipeline handles 10,000 records per hour. What is $(f \circ g)(4)$?

- A.** 16
- B.** 121
- C.** 35
- D.** 176

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A campus IT team surveyed 200 laptops: 120 run antivirus A, 90 have disk encryption, and 45 have both. ~~The survey was run over two weeks by student workers.~~ How many laptops have at least one of the two protections?

Target word: size of the union **Decides:** Subtract the overlap once.

A	Inverted operation	$200 - 165$: laptops with neither.
B	Wrong stem number	The total surveyed.
C	Dropped element	Forgets to subtract the 45 counted twice.
D	Correct	$120 + 90 - 45 = 165$.

2. Answer D Concept 1 · transfer

Sets A and B have $|A| = 40$, $|B| = 35$, and $|A \cup B| = 60$. What is $|A \cap B|$?

Target word: solve for the intersection **Decides:** $|A \cap B| = |A| + |B| - |A \cup B|$.

A	Dropped element	The raw sum, ignoring the union.
B	Inverted operation	Adds the union instead of subtracting it.
C	Confusion set	$60 - 35 = A - B $, elements only in A.
D	Correct	$40 + 35 - 60 = 15$.

3. Answer B Concept 2

A configuration tool offers 6 optional features that can be turned on in any mix, ~~and the vendor charges per seat.~~ How many different feature sets are possible, counting none and all?

Target word: number of subsets **Decides:** Each feature is on or off: 2^6 .

A	Inverted operation	6^2 squares instead of raising 2 to the 6th.
B	Correct	$2^6 = 64$ subsets.
C	Confusion set	$6!$ counts orderings, not subsets.
D	Dropped element	Leaves out the empty set.

4. Answer A Concept 3

A university gives each of its 30,000 students a unique ID drawn from 1,000,000 possible IDs. ~~The ID system was built in 2012.~~ How is the function f : Students $\rightarrow$ IDs best described?

Target word: function type **Decides:** Unique outputs, but most IDs unused.

A	Correct	No two students share an ID, but 970,000 IDs have no student.
B	Confusion set	A function needs one output per input, not full coverage.
C	Inverted operation	Reverses both properties.
D	Adjacent category	Would require every ID to be used.

5. Answer A Concept 4

On the set of devices in a network inventory, define R by “x is on the same VLAN as y.” ~~The inventory lists 812 devices from four vendors.~~ What kind of relation is R?

Target word: classify the relation **Decides:** “Same as” is reflexive, symmetric, and transitive.

A	Correct	Each device shares its own VLAN; sharing is mutual and chains.
B	Dropped element	If x shares a VLAN with y and y with z, x shares it with z.
C	Confusion set	Partial orders are antisymmetric; this relation is symmetric.
D	Camouflage	Vendor does not affect whether R is a relation.

6. Answer C Concept 4 • transfer

The relation $\leq$ on the integers is reflexive, antisymmetric, and transitive. What kind of relation is it?

Target word: reverse: properties to type **Decides:** Reflexive + antisymmetric + transitive = partial order.

A	Confusion set	Needs symmetry, which $\leq$ lacks.
B	Adjacent category	Each integer relates to infinitely many others.
C	Correct	Those three properties define a partial order; every pair is comparable.
D	Absolute language	Relations need not be symmetric.

7. Answer C Concept 5

A help desk closes 40 tickets in week 1 and closes 6 more each week than the week before as new staff ramp up. ~~The manager also tracks satisfaction scores averaging 4.3.~~ How many tickets does it close in week 10?

Target word: 10th term **Decides:** $a_{10} = 40 + 9 \times 6$.

A	Adjacent category	The total for weeks 1-10: $10(40 + 94)/2$.
B	Dropped element	Uses n instead of $n - 1$.
C	Correct	$40 + (10 - 1) \times 6 = 94$.
D	Camouflage	Built from the satisfaction score, which is unrelated.

8. Answer B Concept 5 • transfer

An arithmetic sequence has first term 7 and common difference 5. Which term number equals 102?

Target word: solve for n **Decides:** $7 + (n - 1)5 = 102$.

A	Confusion set	95 is a multiple of 5, so it is a term.
B	Correct	$(102 - 7)/5 = 19 = n - 1$, so $n = 20$.
C	Adjacent category	The distance from the first term, not the term number.
D	Dropped element	Stops at $n - 1$.

9. Answer D Concept 6

Backup storage use starts at 2 TB in year 1 and doubles each year; every year's backups are kept. ~~The array uses RAID 6 across 12 disks.~~ How much storage is used in total across years 1 through 5?

Target word: sum of 5 geometric terms **Decides:** $a_1 = 2$, $r = 2$, $n = 5$.

A	Dropped element	$2 \cdot 2^5$, forgetting the $- 1$ in the sum formula.
B	Inverted operation	Adds 2 each year instead of doubling.
C	Adjacent category	Year 5 alone, not the total.
D	Correct	$2 + 4 + 8 + 16 + 32 = 2(2^5 - 1) = 62$.

10. Answer A Concept 6 · transfer

A geometric sequence has $a_1 = 3$ and $a_4 = 81$. What is the common ratio?

Target word: solve for r **Decides:** $3 \cdot r^3 = 81$.

A	Correct	$r^3 = 27$, so $r = 3$.
B	Dropped element	That is r^3 ; the cube root was skipped.
C	Adjacent category	$(81 - 3)/3$ is an arithmetic difference.
D	Inverted operation	Treats a_4 as $a_1 \cdot r^2$.

11. Answer B Concept 7

A script loops i from 1 to 100 and adds i to a counter on each pass; ~~it runs on a server with 16 cores.~~ What is the final counter value?

Target word: Σi for $i = 1$ to 100 **Decides:** $n(n + 1)/2$.

A	Dropped element	$n^2/2$ drops the $+ n/2$ term.
B	Correct	$100 \times 101 / 2 = 5,050$.
C	Inverted operation	n^2 instead of $n(n + 1)/2$.
D	Camouflage	Multiplies by the core count, which does not change the sum.

12. Answer C Concept 8

In a data-transform pipeline, $f(x) = 2x + 3$ and $g(x) = x^2$; ~~the pipeline handles 10,000 records per hour.~~ What is $(f \circ g)(4)$?

Target word: evaluate a composition **Decides:** Apply g first: $g(4) = 16$.

A	Dropped element	Stops after g .
B	Inverted operation	$(g \circ f)(4) = g(11) = 121$: wrong order.
C	Correct	$f(16) = 2(16) + 3 = 35$.
D	Confusion set	$f(4) \cdot g(4) = 11 \times 16$, a product, not a composition.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Inclusion-exclusion	$ A \cup B = A + B - A \cap B $.	Adding the two sets and double-counting the overlap.	Two overlapping groups and “at least one.”
2	Power sets and products	A set of n elements has 2^n subsets; $ A \times B = A \cdot B $.	Using n^2 , $n!$, or leaving out the empty set.	“How many subsets / combinations of features, including none.”
3	Injective, surjective, bijective	Injective: no two inputs share an output. Surjective: every codomain element is hit. Bijective: both.	Calling a one-to-one map a bijection even when outputs go unused.	Unique assignments from a larger pool.
4	Relation properties	Equivalence = reflexive, symmetric, transitive; partial order = reflexive, antisymmetric, transitive.	Swapping symmetric and antisymmetric.	“Same as” (equivalence) vs. “at most / divides” (order).
5	Arithmetic sequences	$a_n = a_1 + (n - 1)d$; sum of n terms = $n(a_1 + a_n) / 2$.	Using n instead of $n - 1$.	A constant amount added each step.
6	Geometric sequences and series	$a_n = a_1 \cdot r^{(n - 1)}$; sum of n terms = $a_1(r^n - 1)/(r - 1)$.	Adding instead of multiplying, or reporting the last term as the total.	“Doubles” or “grows by a factor.”
7	Summation formulas	$\sum_{i=1}^n i = n(n + 1)/2$.	Using n^2 or $n^2/2$.	A loop adding the counter itself.
8	Function composition	$(f \circ g)(x) = f(g(x))$: apply g first, then f .	Applying f first, or multiplying f and g .	“ $f \circ g$ ” or “pipeline of transforms.”

Course scope (public description): Discrete Math: Functions and Relations is designed to help students develop competence in the use of abstract discrete structures fundamental to systems networking. In particular, this course will introduce students to set theory, finite sequences, series, and relations. Discrete Math: Logic, Applied Probability and Statistics, and Applied Algebra are prerequisites for this course.